

1.80

DP on Grids / 2D Matrix

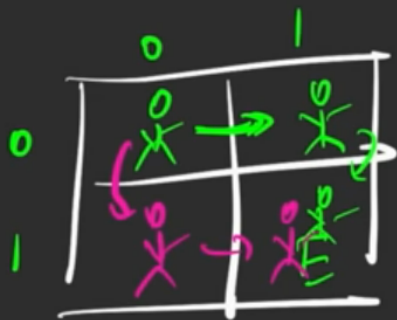
- Count Paths
- Count Paths with obstacles
- Min Path Sum
- Max Path Sum
- Triangle
- 2 start points

6 problems

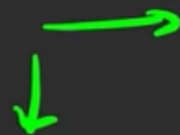


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Unique Paths



2 x 2



(0,0) → (1,1)

2 unique ways



How to write Recurrence ?

(2D grid)
row col
(i, j)

1. Express in terms of LA
2. Explore / Do all stuffs
3. Sum up all weights | Max / Min



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$f(m-1, n-1)$

$f(i, j) \rightarrow$ no. of unique ways
 $(0, 0) \rightarrow (i, j)$

$(0, 0) \rightarrow (m-1, n-1)$



$f(i, j)$



$f(i, j)$

{
if ($i == 0$ || $j == 0$)

return 1;

if ($i < 0$ || $j < 0$) return 0;

up = $f(i-1, j)$;

left = $f(i, j-1)$;

return $\boxed{\text{up} + \text{left}}$;

}

Base Cases

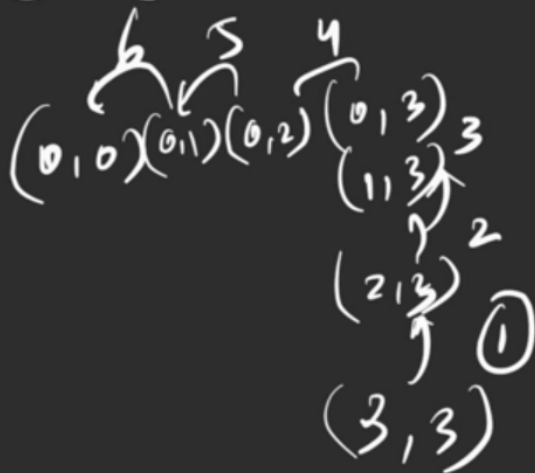
↑ up
← left



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Recursion

TC →



TC → $(2^{m \times n})$

SC → O (path length)

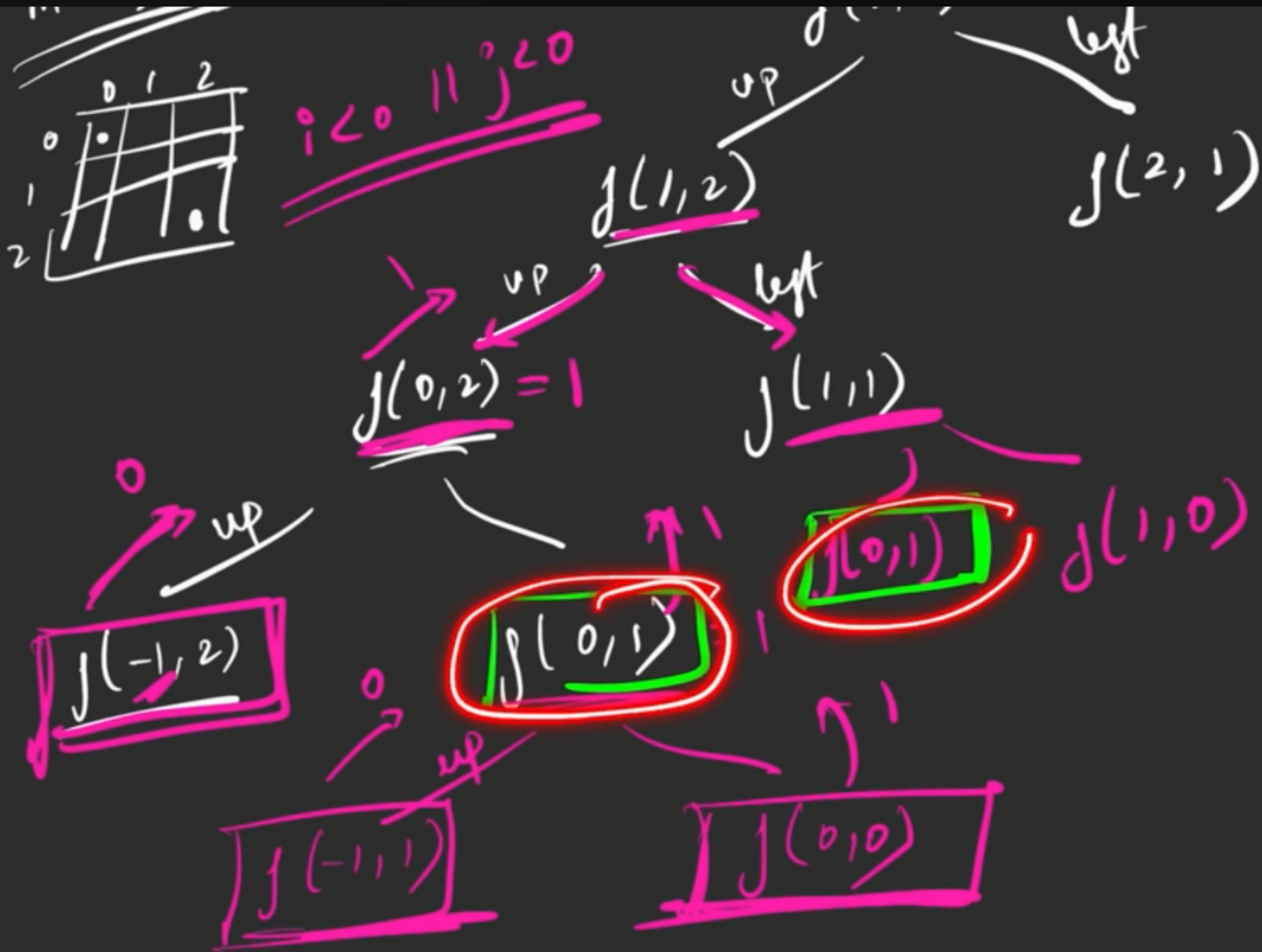
✓
(m-1) + (n-1)



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Recursion → DP
↳ Memorization
Overlapping
(Subproblems)





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Memor → Tabulation

1. Declare Base Case
2. Traverse ^{all} various states in for loop
3. Copy the recurrence & write.



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dp[0][0] = 1

for (i = 0 → m-1)

{
 for (j = 0 → n-1)

{
 if (i == 0 & j == 0) dp[0][0] = 1

else {

 up = dp[i-1][j];

 left = dp[i][j-1];

 dp[i][j] = up + left;

}

}

}



dp[0][0] = 1

for (i = 0 → m-1)

{
 for (j = 0 → n-1)

 { if (i == 0 & j == 0) dp[0][0] = 1

 else {

 if (i > 0) up = dp[i-1][j];

 if (j > 0) left = dp[i][j-1];

 dp[i][j] = up + left;

 }

}

}



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$$dp[i][j] = \underbrace{dp[i-1][j]}_{up} + \underbrace{dp[i][j-1]}_{left}$$

~~dp[i][j]~~ →
~~temp~~ → dp →
~~temp~~ →

0	0	0	0	0
1	1	1	1	1
2	2	3	3	4

$i=0$
 $j=0$



1D array

$dp[n] \rightarrow 0$

for ($i = 0 \rightarrow n-1$)
{
 temp[n] = 0

for ($j = 0 \rightarrow n-1$)

{
 if ($i == 0$ && $j == 0$) $dp[i][j] = 1$;
 else {

temp[j] =

$dp[j] +$
temp[j-1]



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```

// dp [n][n]
for (i = 0 -> n-1)
    temp[n] = 0
for (j = 0 -> n-1)
    if (i == 0 && j == 0) dp[i][j] = 1;
    else {
        temp[j] = dp[i][j] + temp[j-1];
        prev = temp;
    }
    }
    )
    
```



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Problem Submissions

You don't have to print anything, it has already been taken care of. Just implement the given function.

Constraints:

```
1 ≤ T ≤ 100
1 ≤ M ≤ 15
1 ≤ N ≤ 15
```

Where 'M' is the number of rows and 'N' is the number of columns in the matrix.

Time limit: 1 sec

Sample Input 1:

```
2
2 2
1 1
```

Sample Output 1:

```
1 int f(int i, int j) {
2     if(i == 0 && j == 0) return 1;
3     if(i < 0 || j < 0) return 0;
4     int up = f(i-1, j);
5     int left = f(i, j-1);
6     return up + left;
7 }
8 int uniquePaths(int m, int n) {
9     return f(m-1, n-1);
10 }
```

Sample Test Case Custom Test Case

Show Hint

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Problem

Submissions



Time Limit
Exceeded



Show
Hint



Report

(Partially Accepted)

Submitted on Jan 18, 2022, 5:57:43 PM



Your time complexity: $O(m*n)$ - For 2d arrays

We think Common causes of Time Limit Exceeded :

Here is what went wrong with your code

Penalty

0%

Score ?

91

Runtime

-

Language

C++ (g++ 5.4)

```
1 int f(int i, int j, vector<vector<int>> &dp) {
2     if(i == 0 && j == 0) return 1;
3     if(i < 0 || j < 0) return 0;
4     if(dp[i][j] != -1) return dp[i][j];
5     int up = f(i-1, j, dp);
6     int left = f(i, j-1, dp);
7     return dp[i][j] = up + left;
8 }
9 int uniquePaths(int m, int n) {
10     vector<vector<int>> dp(m, vector<int>(n, -1));
11     return f(m-1, n-1, dp);
12 }
```

Sample Test Case

Custom Test Case

Test Case 1



Correct Answer

Test Case 2



Correct Answer

Test Case 3

Correct Answer

Test Case 4



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Time Limit
Exceeded

(Partially Accepted)

Submitted on Jan 18, 2022, 5:57:43 PM



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Hint



Report



Your time complexity: $O(m*n)$ - For 2d arrays

We think Common causes of Time Limit Exceeded :

Here is what went wrong with your code

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Score ?

91

Runtime

-

Language

C++ (g++ 5.4)

```
4     if(dp[i][j] != -1) return dp[i][j];
5     int up = f(i-1, j, dp);
6     int left = f(i, j-1, dp);
7     return dp[i][j] = up + left;
8 }
9 int uniquePaths(int m, int n) {
10    int dp[m][n];
11    for(int i = 0; i < m; i++) {
12        for(int j = 0; j < n; j++) {
13            if(i == 0 && j == 0) dp[i][j] = 1;
14            else {
15                int up = 0;
16                int left = 0;
17                if(i > 0) up = dp[i-1][j];
18                if(j > 0) left = dp[i][j-1];
19                dp[i][j] = up + left;
20            }
21        }
22    }
23    return dp[m-1][n-1];
24 }
```



Previous

Next



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44:50 / 48:28

Code implementation and wrap-up >



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Problem

Submissions



Time Limit

Exceeded

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Hint



Report

⚠️ Your time complexity: $O(m*n)$ - For 2d arrays

We think Common causes of Time Limit Exceeded :

- Your time complexity: $O(m*n)$ - For 2d arrays
- Think of some large use-cases and try again!

Found this helpful? 👍 👎

Penalty

0%

Score ?

91

Runtime

-

Language

C++ (g++ 5.4)

Previous Submissions

Submitted On

Status

Score

Penalty



Previous

Next



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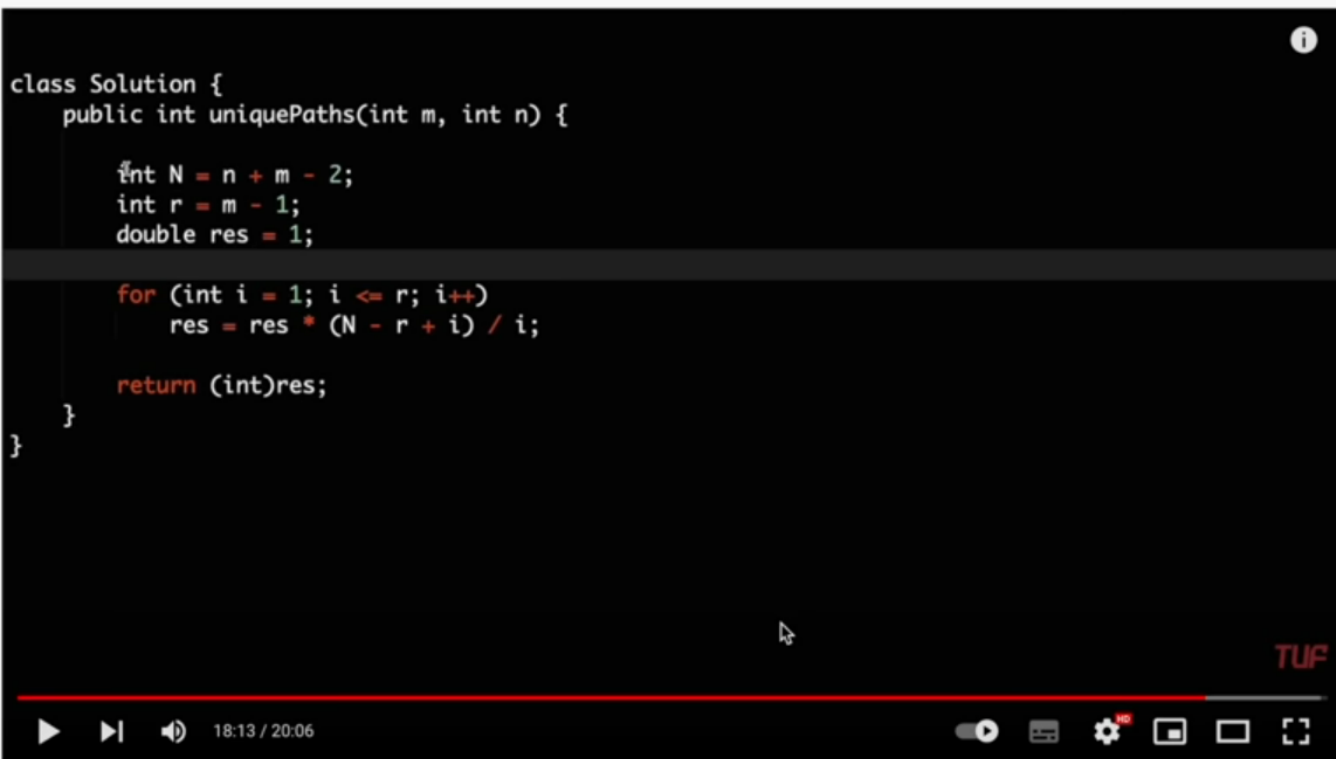
```
6     int left = f(i, j-1, dp);
7     return dp[i][j] = up + left;
8 }
9 int uniquePaths(int m, int n) {
10     vector<int> prev(n, 0);
11     for(int i = 0; i < m; i++) {
12         vector<int> cur(n, 0);
13         for(int j = 0; j < n; j++) {
14             if(i == 0 && j == 0) cur[j] = 1;
15             else {
16                 int up = 0;
17                 int left = 0;
18                 if(i > 0) up = prev[j];
19                 if(j > 0) left = cur[j-1];
20                 dp[i][j] = up + left;
21             }
22         }
23         prev = cur;
24     }
25     return prev[n-1];
26 }
```



46:39 / 48:28

Code implementation and wrap-up >





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